

Liangyu WU | Curriculum Vitae

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Education Background

Stanford University <i>Ph.D. Candidate in Physics, Physics Department</i> (Advisor: Dr. Julia Gonski & Dr. Dong Su) • Research area: Experimental Particle Physics	California, U.S. Sep. 2024-Present
Shanghai Jiao Tong University <i>Bachelor of Science in Physics, School of Physics and Astronomy</i> • Coursework: <i>Quantum Mechanics, Electrodynamics, Equations of Mathematical Physics, Professional Experiment, Introduction to Nuclear and Particle Physics, Methods of Experimental Nuclear and Particle Physics</i> (All graded A's)	Shanghai, China Sep. 2020- Jun. 2024
University of Maryland, College Park <i>Visiting International Student</i>	Maryland, U.S. Aug. 2023-Dec. 2023

Research Experience

SLAC National Accelerator Laboratory <i>Particle Physics and Astrophysics</i> <i>Research Assistant</i> (Supervisor: Dr. Julia Gonski & Dr. Ariel Schwartzman & Dr. Dong Su) – Research on the Future Colliders • Development of real-time machine learning algorithms for deployment on front-end electronics in future detectors. • R&D of embedded FPGA (eFPGA) technology for ultra-low latency signal processing applications in future colliders. • Investigation of novel digital sensor technologies (e.g., digital SiPMs) for calorimetry readout in future colliders. – Research on the ATLAS Experiment at CERN • Testing of custom front-end ASICs for high-speed data transmission in the ATLAS ITk upgrade for HL-LHC. • Advancing LAr calorimeter timing to extend fast-timing detector coverage beyond HGTD. • Development of novel machine learning-based foundation models for new physics searches.	California, U.S. Jan. 2025-Present
SLAC National Accelerator Laboratory <i>Particle Physics and Astrophysics</i> <i>Research Assistant</i> (Supervisor: Prof. Spencer Gessner) – Research on the FACET-II • Development of user interface software and data analysis tools for advanced accelerator systems.	California, U.S. Sep. 2024-Dec. 2024
University of Maryland, College Park <i>Department of Physics</i> <i>Undergraduate Research Student</i> (Supervisor: Prof. Christopher Palmer & Prof. Sarah Eno) – Research on the Dual-Readout Calorimetry • Monte Carlo simulation studies of detector response optimization across diverse calorimeters. • Mathematical modeling and algorithm development for multi-channel signal processing and calibration.	Maryland, U.S. Aug. 2023-Aug. 2024
Shanghai Jiao Tong University <i>School of Physics and Astronomy</i> <i>Undergraduate Research Assistant</i> (Supervisor: Prof. Yue Meng) – R&D for a Novel Radon Detector • Design and construction of novel detector with precision measurement capabilities for material characterization.	Shanghai, China Oct. 2021-Feb. 2024

– Work in Ultra-low background technique R&D group

- Material characterization and ultra-low background technique implementation for precision detection systems.

– Work in PMT group

- Testing, optimization, and installation of high-voltage detection systems and power distribution infrastructure.

Tsung-Dao Lee Institute

Undergraduate Research Student (Supervisor: **Prof. Masahiro Ogihara**)

– Research on the stability of unevenly spaced planetary systems

- N -body computational simulations for dynamic system stability analysis.
- Statistical modeling demonstrating systematic biases in conventional analytical approaches.

Publications

Instrumentation & Detector R&D

- [1] **Machine Learning Enables Real-Time Waveform Decomposition for Dual-Readout Calorimetry**, L. Wu, Q. Liu, M. T. Lucchini, J. Gonski, M. Campajola, S. Moneta, [arXiv:2604.26090](#)
- [2] **Agentic-AI Detector Co-design and Optimization in Vertically-Integrated Differentiable Full Simulations**, W. Chung, Q. Liu, L. Wu, J. Gonski, [Accepted to Stanford Physics & AI Conference 2026](#)
- [3] **Edge Machine Learning for Cluster Counting in Next-Generation Drift Chambers**, D. Yilmaz, L. Wu et al., [Accepted by NeurIPS 2025](#)
- [4] **On the resolution of dual readout calorimeters**, S. Eno, L. Wu et al., [NIM A](#), Volume 1083, 2026, 171080, ISSN 0168-9002
- [5] **Design and Experimental Application of a Radon Diffusion Chamber for Determining Diffusion Coefficients in Membrane Materials**, L. Wu, L. Si et al., [JINST](#), 20, no.03, P03031 (2025)
- [6] **Geant4 simulations of sampling and homogeneous hadronic calorimeters with dual readout for future colliders**, S.V. Chekanov, S. Eno, S. Magill, C. Palmer, L. Wu, [NIM A](#), Volume 1072, 2025, 170200, ISSN 0168-9002

Physics Analysis

- [1] **Towards anomaly detection searches for new physics signatures including Higgs bosons with weakly supervised machine learning**, C. L. Cheng, J. Gonski, R. Li, Q. Liu, B. Nachman, D. Noll, J. K. Romman, L. Wu, [Submitted to JHEP](#)

Other

- [1] **A Lightweight Foundation Model for Collider Physics with Multi-Domain Adaptation**, L. Wu, Q. Liu, A. Yue, J. Gonski, [arXiv:2607.27501](#)
- [2] **The stability of unevenly spaced planetary systems**, S. Yang, L. Wu et al., [ICARUS](#), Volume 406, December 2023

Conference Talks & Posters

Conference Talks

- 1. [Mar 2026] **ITk Week 2026**
 - [GBCR Testing](#)
- 2. [Dec 2025] **ATLAS Timing Workshop**
 - [Vertex \$t_0\$ Reconstruction Using LAr Calorimeter with Outlook for LAr-HGTD Combination](#)
- 3. [Oct 2025] **Coordinating Panel on Advanced Detectors (CPAD) 2025**
 - [Developing Digital Silicon Photomultiplier \(dSiPM\) Specifications for a High-Granularity Calorimeter with Simulations](#)
 - [Embedded ML Solutions for Real-time Processing in Future Drift Chambers](#)
- 4. [Jul 2025] **US ATLAS Summer Workshop 2025**
 - [Enhancing Timing Information Reconstruction with LAr Calorimeter](#)
- 5. [May 2025] **ITk Week 2025**
 - [Recent Results of GBCR3 Testing at SLAC](#)

Posters

- 1. [Jun 2026] **ATLAS Week 2026**
 - *The GBCR ASIC for ATLAS ITk Pixel Readout: Testing Framework and Performance Validation*
- 2. [Jun 2026] **FCC Week 2026**
 - *Machine Learning Enables Real-Time Waveform Decomposition for Dual-Readout Calorimetry*

Skills

Technical: C++, Python, CERN ROOT, Geant4, SolidWorks, COMSOL, LaTeX
Hardware: Silicon microelectronics, Printed circuit boards (I/O, transmission, debugging), High-voltage technology
Language: English (Fluent), Mandarin (Native)

Extra-Curricular Experience

Student Union of School of Physics and Astronomy	Shanghai, China
<i>Co-President of the Student Union</i>	Dec. 2021-Dec. 2022
– Spearheaded the execution of events and promotions for the Student Union.	

Selected Honors and Awards

2026 Early Career Professional Award, Future Circular Collider (FCC) Week
2024 Shanghai Outstanding Graduate honors, Shanghai Education Commission